FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

SJ

Singha Junchan

สิงหา จุลจันทร์ / Pathum Thani, Thailand

I build access control for cloud and AI systems.

- **Best Paper Award**, IEEE JCSSE 2026, for **U-HAP**, Kubernetes authorization (second author).
- **9 security research projects** on **Kubernetes**, **zero trust**, **zero-knowledge proofs**, **post-quantum** cryptography and **AI agents**; first author of **FEDAC** (under review, IEEE TDSC).
- **Security intern, Allianz Ayudhya**: learned **security governance and risk** from the CISO and built **two AI agents for third-party risk**.
- **Tech lead, TU Alert**: campus incident reporting in **LINE**, built for Thammasat University (launching soon); I lead the team, design the architecture and the security, and write code.

Scan or click for the full portfolio, code links and contact.

singhaju.github.io

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github.com/singhaju

Open to: security internships from January 2027, in Thailand or abroad full-time roles from May 2027

COMPETITIVE PROGRAMMING

#1 in Thailand

TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT

TU Alert

built for **Thammasat University**; **tech lead** (architecture, security, code); security built in; launching soon

RESEARCH

9 research projects

2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY

2 internships

Allianz Ayudhya (security governance and risk, with the CISO) and **Amity Solutions** (LLM chatbots)

LEADERSHIP AND TEACHING

11 student events

led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION

ISC2 Certified

Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 Experience

knowledge | tech used

Tech Lead and Developer, TU Alert

Launching soon 2026

Thammasat University (client project) / Rangsit campus, Pathum Thani

- **What it is:** a campus incident-reporting system that runs **inside LINE**, built for **Thammasat University** (launching soon). A student reports a problem with photos and a map pin, and a rule sends it to the office that owns it. As **tech lead** I lead the team, design the architecture and the security, and write code.
- **LINE integration:** a student **LINE Official Account** (rich menu, Flex message cards, chat commands, star ratings by Quick Reply), a **group bot** that alerts each office's LINE group and answers /queue and /summary, and two **LIFF apps** with LINE sign-in checked on the server.
- **Student side:** report with photo, video and a campus map pin (kept on the phone until sent), track the status, chat with the office, rate the result, reopen it if not fixed. **Staff side:** a phone app and a web portal to accept, transfer, resolve and assign; SLA and analytics for heads; emergency broadcast to students.
- **Security built in:** four roles checked on every request, second-step staff sign-in (authenticator app, SMS or LINE approval), kill switches, an audit log, PDPA data export and erasure; **Semgrep, gitleaks and OSV** scans in CI.

System architecture | Secure web development | Role-based access control | Team leadership | TypeScript | Next.js | Hono | Bun | Prisma | MariaDB | Redis | LINE LIFF | Semgrep | gitleaks

Information Security Intern

Jun to Jul 2026

Allianz Ayudhya / Bangkok, Thailand

- **What it is:** the Information Security team's **third-party risk management (TPRM)**: checking a supplier's security when the company buys, and answering clients' **due-diligence questionnaires** when it supplies. **Mentored by the CISO**, I learned how **security governance and risk management** work in practice.
- **Designing AI into the work:** mapped the manual questionnaire process end to end with the team to find where AI helps most, and set what every AI answer must carry: **evidence, a reason and a confidence score**, so a person can check it fast.
- **Questionnaire agent:** reads policy and evidence documents, drafts each answer, and runs **gap and fit-gap analysis** to show where current controls fall short. **Guidance agent:** teaches business owners each step, tracks their progress, checks their inputs, **suggests what to do next and flags** items to review.
- Tested on **real, anonymized cases** with the team; demonstrated on 6 Jul and **presented both** on 31 Jul 2026.

Information security governance | Risk management | Third-party risk management | Due-diligence questionnaires | Gap and fit-gap analysis | AI agents

01 Experience

continued

Founder, Curriculum Author and Lead Instructor

TOI Tutor and ABIG Center (online) / toitutor.in.th — 2022 to present

- **What it is:** two online platforms that coach school students for Thailand's national programming olympiads. **Founded and run** since 2022; **coached hundreds of students** in live classes and one to one.
- **Courses I wrote and teach: COM300**, C++ and algorithms (9 chapters, about 270 pages, **128 recorded lessons**), and **COM100**, maths and **Python** for the olympiad exam (10 chapters with answer keys).

Algorithms and data structures · Competitive programming · Curriculum design · Teaching · C++ · Python

R&D Intern, AI research team

Amity Solutions / Bangkok, Thailand — Jul to Aug 2023

- **What it is:** the AI research and development team, building chatbots on **large language models (LLMs)**.
- **From paper to pipeline:** read recent **LLM research papers**, applied their ideas in the team's pipeline and **tested them end to end**, from the user's question to the answer.
- **RAG chatbots:** take the question, **search the knowledge base**, give the best-matching documents to the LLM with the question, and return its answer. Also data analysis and **prompt engineering**.

Large language models · Retrieval-augmented generation (RAG) · Applying research papers · End-to-end testing · Prompt engineering · Data analysis

02 Research

Research supervisor: Prof. Somchart Fugkeaw, School of ICT, SIIT.

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8. One question links most of them: can a security check be cheap enough to run on every request, and still be provable afterwards?

knowledge · tech used

FEDAC

● Under review · IEEE TDSC

ZERO-TRUST ACCESS CONTROL — 1st of 4 authors

In Kubernetes systems spread over many clusters and organisations, checking every request against the access rules is slow, and the rules are hard to trust. FEDAC checks **once** whether a caller may use a service, then gives a **short-lived pass for that one service** that is checked quickly on every call.

- Sign-in gets **stronger as the service gets more sensitive**, up to a zero-knowledge proof, and a **risk score** (device, location, behaviour) can stop a risky sign-in.
- **Different kinds of access rule** are turned into one common format, so one system enforces them all.
- A user shows only the details a service needs, so services **cannot track the user** across each other.
- Rules are stored so anyone can **check they were not changed** (IPFS and a blockchain) without slowing requests; when a rule changes, only the affected passes are cancelled.

Zero trust · Zero-knowledge proofs · Service mesh · Blockchain audit · Cloud benchmarking · Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

U-HAP

● Presented · IEEE JCSSE 2026 · Best Paper Award

ZERO-TRUST ACCESS CONTROL — 2nd of 4 authors · presented the paper

Kubernetes has several kinds of access rule (RBAC, ABAC, ACL, deny rules), handled by separate parts that do not know about each other. U-HAP puts **all of them into one graph** and prepares it ahead of time, so each request is answered with a **quick lookup**.

- **One meaning for every rule type**, so existing policies do not need to be rewritten.
- Rules are **prepared before requests arrive**, so a request never scans the whole rule set.
- Conflicts follow clear rules (**a deny always wins**), and its decisions were checked against Open Policy Agent.

Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Graph models · Python · Kubernetes webhook · OPA (baseline)

ZKAudit

● Accepted · KST 2027

ZERO-KNOWLEDGE AUDIT — First author

Companies let outside auditors check their cloud logs. ZKAudit lets an auditor **prove it is allowed to audit without revealing its credential**, and shows that the **logs were not changed**.

- The proof (a **zero-knowledge proof**) is tied to the exact logs the auditor then checks, so the permission cannot be used for anything else.
- Proof checking is **shared across independent verifiers**, so many auditors can be admitted at once with no single bottleneck.
- Logs are chained with **Merkle trees** on an append-only ledger, so tampering shows and one record can be checked without reading the whole history.

Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Privacy · Rust · Groth16 zk-SNARKs · AWS EC2

Post-quantum authorization across Kubernetes clusters

Submitted

POST-QUANTUM SECURITY

First author

When a request moves between Kubernetes clusters, the target cluster usually has to trust the other side's access decision. Here **each cluster checks the decision for itself**, with signatures built to resist future **quantum computers**.

- The proof shows that **every allow and deny rule** was considered, so a deny rule cannot be quietly left out.
- A role from one cluster **cannot silently become a stronger role** in another; the target cluster approves the mapping.
- The heavy post-quantum check is done **once per session**, then lighter checks follow; when a rule changes, only the affected decisions are cancelled.

Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Distributed ledgers · Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

Self-healing fog computing for industrial IoT

Submitted

RESILIENT CLOUD AND EDGE

First author

Fog computing runs work on small servers close to factory machines. Here the clusters **cooperate like a swarm, with no central controller**: they share work and **recover from failures by themselves**.

- Clusters leave **small signals for each other** about load and health, like ant trails, and use them to decide where work goes.
- **Backups and recovery plans are ready before anything fails**, so a failed coordinator is replaced quickly.
- Clusters **learn good strategies from neighbours**; shared data is protected with homomorphic encryption and Merkle proofs.

Fog and edge computing · Industrial IoT · Fault tolerance · Homomorphic encryption · Python · gRPC · Paillier · Merkle proofs · AWS EC2

MoCA: AI containment

In preparation

AI-AGENT SECURITY

First author

AI agents can be tricked by hidden instructions in what they read (**prompt injection**). MoCA assumes this will sometimes work, and makes sure the agent **still cannot do more than the user allowed**.

- Permissions are checked **when the agent acts**, not when it plans.
- An agent **never gets more rights than the user** who sent it, even when it hands work to other agents.
- Tools must be **approved**; a changed tool stops working until approved again, and checks run at a gateway and at the resource.

AI-agent security · Delegation and revocation · Prompt injection · Python · MCP tool gateway · Attack test harness

ZK-ALBA

In preparation · journal paper

ZERO-KNOWLEDGE AUDIT

First author

Extends ZKAudit: an auditor proves it may audit **right now**, and checks that the logs it receives are **complete and untouched**, while the logs **stay encrypted**.

- Each proof is tied to **one audit, one set of logs and one auditor**, so it cannot be reused.
- The auditor checks the encrypted logs **without seeing their content or the keys**.

Zero-knowledge proofs · Scoped audit · Load balancing · Replay protection · Rust · Sharded verifiers · AWS EC2

Threshold-gated encrypted SIEM

In preparation

SECURITY OPERATIONS

First author

An attack can move across several companies, but no company wants to share readable logs. This system **follows the attack across companies' encrypted logs**, and opens a trail into another company only when **several companies independently saw the same warning sign**.

- Logs stay **encrypted under each company's own key**.
- The platform learns only that enough companies matched, **never which ones**.

SIEM and threat detection · Applied cryptography · Threshold schemes · Privacy across companies · Python · Detection pipeline · LANL enterprise dataset

02 Research

continued

Fresh answers from elastic multi-cloud services

In preparation

RESILIENT CLOUD AND EDGE

First author

When a service that keeps data grows across several clouds, a new or distant copy can **serve old data**. This scheduler sends requests only to copies that **prove they are up to date**.

- Small **Merkle proofs** show that a copy is up to date, with no blockchain in the way.
- Among up-to-date copies it picks by data location, network distance, queue and GPUs; a new copy serves **only after it has caught up**.
- After a failure it **moves traffic to an up-to-date copy** instead of a slow restart, when that is cheaper.

Multi-cloud systems · Elastic scaling · Freshness proofs · Python · Sparse Merkle trees · Multi-cloud deployment

How I work on each project

1. Find the gap in published designs
2. Design the scheme
3. Build the prototype
4. Measure it on AWS or real data
5. Write the paper

03 Projects

TU Alert

TypeScript

No public repo

Campus incident reporting **inside LINE**, built for Thammasat University (launching soon): students report with photos and a map pin, and each report goes to the office that owns it. **Tech lead**: I lead the team, design the architecture and the security, and write code.

web · full-stack · security · team

Novel Nest

TypeScript · MySQL

Public repository

A web platform where readers browse and review serialized novels and writers publish episodes (Next.js, MySQL). **Project manager and product owner** of a team of four; I also wrote full-stack code (sign-in, reviews, admin and episode pages).

full-stack · team

Arduino retro console

C++ · Arduino

Public repository

A handheld game console on an Arduino with a small OLED screen, running Snake, Flappy Bird and Tetris. **I wrote the menu, controls and drawing loop** and joined the games together (games adapted from open-source projects).

embedded

Automated Cat Feeder

Node-RED · C++ · SOLIDWORKS

No public repo

An IoT cat feeder: 3D-printed prototype, feeding from the web, schedules and a live camera. **I built the MQTT link** between the Raspberry Pi and the web app. **Bronze Medal and Popular Vote**, SCIUS Forum 2022.

IoT · award

Velour Mall

Figma · Canva

Public repository

A **UX/UI design** case study in Figma for a shopping-mall app: indoor navigation, a smart parking assistant and an AI chatbot.

design

04 Hands-on security

NCSA CTF Thailand Tournament 2026

Thailand Cyber Top Talent 2026, Aug 2026. Competed in a team on web, cryptography, digital forensics, OSINT and mobile challenges.

SOSECURE Road to Cybersecurity 2025 (GEN6)

Eight modules with labs: routing and firewalls, Linux hardening, Python security tools, Metasploit and web attacks, SOC log analysis in Splunk and Wazuh.

Ongoing practice

TryHackMe and Hack The Box. Nmap scanning in a Kali Linux home lab. DataCamp courses on Azure and AWS (2025).

05 Education and leadership

Education

B.Eng. in Computer Engineering — Aug 2023 to May 2027 (expected)
Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University GPA 3.67 / 4.00, first-class honours expected.

Leadership

11 student events organised · ~500 participants at SIIT First Meet 2024 · 100 staff at SIIT First Meet 2024

President, Sub Student Committee 32, SIIT — 2024 to present
Leads a committee that represents the student body and organises academic and social events.

Lead organiser, SIIT First Meet 2024 — 2024
About 500 participants and 100 staff. Organiser of 10 more student events, 2023 to 2025.

Project manager and product owner, Novel Nest — Team project
Led a team of four to build a digital library and creative hub for readers and writers.

06 Awards and certifications

2022 — **National Champion: 1st place in Thailand and Gold Medal**
Thailand Olympiad in Informatics (TOI18)

2025 — **1st place overall**
Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

2022 — **Bronze Medal and Popular Vote**
SCIUS Forum 2022, Automated Cat Feeder

2026 — **Best Paper Award**
IEEE JCSSE 2026, Bangkok, for U-HAP (2nd of 4 authors)

2023 to 2027 — **Full scholarship**
4-Year Continuing Academic Outstanding Student Program, SIIT

CERTIFICATIONS ✓ **ISC2 Certified in Cybersecurity (CC)** 2025 planned **CompTIA Security+** Exam planned Dec 2026

07 Skills

Programming	**Python** · **C++** (main) · Go · Rust · TypeScript · C · Java · Bash · SQL
Used in my research	Zero Trust · Kubernetes (RBAC, ABAC, webhooks) · AWS (EKS, EC2) · Docker · Envoy and Istio · OPA / Rego · IPFS · Hyperledger Fabric
AI agents and AI security	LLM agent pipelines · MCP gateway · Prompt-injection testing · Prompt engineering · NLP chatbots
Shipping and operations	GitHub Actions CI · Semgrep · gitleaks · Playwright · pm2 on Linux · Backups and restore
Cryptography	zk-SNARKs (Groth16) · Merkle commitments · ML-DSA · ML-KEM
Security tools (labs and CTFs)	Nmap · Wireshark · Burp Suite · Metasploit · OSINT · OWASP Top 10
Also familiar with	Splunk and Wazuh (SOC labs) · Azure and AWS basics (DataCamp)
Languages	Thai (native) · English (fluent) · Chinese (basic)

08 References

People I have worked with, who agreed to be named. Contact details on request.

Apichai Wongchindakhun
Chief Information Security Officer (CISO)
Allianz Ayudhya
Mentored me during my information security internship (Jun to Jul 2026), where I built and presented two AI-agent prototypes for third-party risk.
Internship mentor

Prof. Somchart Fugkeaw
Professor, School of ICT
SIIT, Thammasat University
Supervises my research and co-authors my papers, including FEDAC (under review, IEEE TDSC) and ZKAudit (accepted, KST 2027).
Research supervisor

Pachara
Co-developer, TU Alert
SIIT, Thammasat University
Builds TU Alert with me for Thammasat University.
Project partner

Let's talk
Looking for an internship from January 2027 and full-time roles from May 2027: security engineering, cloud security or AI security.

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